

When to audit a free claim: dispatching a scarce mobile actuator under strategic urgency reports

Hao-Ting (Lex) Ho

Working draft, not peer reviewed — October 2026

Abstract

A single mobile actuator must restore the perception health of N explorers on a ring. Explorers’ health decays with hidden shocks, they report a noisy urgency for free, and travel and non-interruptible service make every dispatch costly. We use exact per-step dynamic programming ($N \leq 4$) and a GPU simulator to ask when reports are worth anything and what happens when they are strategic. In the cell $N=3$, shock fraction 0.5, spread 0.6, service 5, the exact value of free perfect state information is $\text{VoI}_{\text{step}} = 0.012/0.019/0.023/0.018$ team-health units at load $r = .002/.004/.008/.016$ (K-extrapolated, 32 seeds); it rises monotonically with the shock fraction. We do not claim that this is below the pre-specified (internal protocol) threshold of 0.02. Naive trust in reports (**fused_index**) loses (team-health units, truthful minus approximate best response) 0.247 [0.223, 0.272] at $N=3$, $r=.004$, shock fraction .5 (57 actions) and 0.40 [0.36, 0.44] at $N=8$, $r=.002$, shock fraction .5 (24 actions) against an approximate best-response profile; this is a quantitative instance of the cheap-talk collapse, not a new phenomenon. Auditing at arrival is bypassed by distance-dependent inflation (loss 0.024 [0.010, 0.038] at $N=3$; 0.042 [0.025, 0.059] at $N=8$). Auditing at arrival clearly fails: the loss CI excludes 0 in 9 of the 10 cells run. Auditing at dispatch has a smaller loss within the tested strategy families and a limited best-response search (0.0027 [0.0001, 0.0052] and 0.0058 [0.002, 0.010] in the same two cells, about 1/7–1/9 of the arrival audit), but the evidence is inconsistent: the CI contains 0 in 5 of the 10 cells (no multiple-comparison correction) and best-response convergence is low (0.25 at $N=3$, 0.00 at $N=8$). Because a search that has not converged may simply miss profitable deviations, the loss may be underestimated and the ratio should be read as an optimistic estimate for the dispatch audit; our pre-specified claim that the loss vanishes was refuted and is withdrawn. Pricing requests by distance, an exploratory result, mostly acts as a crude distance filter. All numbers refer to a reduced experimental grid (10 of 27 pre-specified cells), disclosed in Section 8.

1 Introduction

Consider a multi-robot exploration team with one actuator (a “super-robot”) that can restore a degraded explorer’s perception. Explorer B calls for help, the actuator sets off toward B , then explorer A calls. Where should the actuator go? The folk answer is *serve the most urgent*. We ask when that is right if urgency is private, noisy and free to claim, and if reaching a robot costs travel time.

This paper is the corrected continuation (analysis plan written before the final confirmatory runs) of an earlier exploratory repository whose headline was that “information has no value” and that “pricing the request” recovers the loss. Both statements were weakened or reversed by later analysis (Section 8); we report what survived. Our contributions, in the order of the evidence strength:

1. **Exact value of information with travel coupling.** Exact per-step dynamic programming (DP) at $N = 3$ gives the value of free perfect state information, VoI_{step} , as a function of load and (at one load) of shock fraction and heterogeneity (Sections 4–5).
2. **Audit timing.** In a reporting game, audit at arrival is bypassed by distance-dependent inflation; the arrival audit fails clearly (loss CI excludes 0 in 9 of 10 cells), whereas the dispatch audit has a smaller loss within the tested strategy families and a limited best-response search (about 1/7–1/9 of the arrival audit in the same cell), but with inconsistent evidence (CI contains 0 in 5 of 10 cells, uncorrected for multiple

comparisons; convergence 0.25 at $N=3$, 0.00 at $N=8$). Since non-converged search may miss profitable deviations, the ratio is an optimistic estimate for the dispatch audit (Section 6).

3. **Quantitative checks of known theory (not new phenomena).** Naive trust collapses as cheap-talk theory predicts (C4); a request price λd behaves like a Naor-type toll but is only a coarse distance filter (C7); myopic urgency-first dispatch is dominated by touring and distance-aware rules as in the dynamic traveling repairman problem (C1).

Everything here rests on a stylised ring model, finite strategy families, and a reduced grid; Section 9 lists what we cannot claim.

2 Related work

Dynamic traveling repairman and polling. Serve-nearest and partition-plus-tour policies beat FCFS-type disciplines under load [3, 4]; see Bullo et al. [6] for robotics. In polling systems with switchover times the server is not work-conserving and only pseudo-conservation laws hold [5, 18]. Our round-robin baseline is a polling system with switchover equal to travel; our finding that urgency-first dispatch loses to touring reproduces this literature qualitatively and we treat it as a simulator sanity check.

Restless bandits and switching cost. The problem is a restless-bandit-like allocation [23, 22]; closed-form indices exist for reset-on-service maintenance and surveillance arms without travel [10, 21], and switching-cost index results cover non-restless arms [1], and for restless arms with general switching costs only heuristics are known [14]. A mobile repairman on a network is treated with index heuristics and rollout by Tian and Shone [19]. We did not find a published index for decaying arms with position-dependent travel (a negative search, not proof of absence); we do not attempt a Whittle index, because the travel coupling breaks per-arm decomposition, and benchmark instead by exact DP at $N \leq 4$.

Cheap talk and artificial currencies. In cheap-talk models free, unverifiable messages convey only coarse information when interests diverge, and a babbling outcome always exists [7]; repeated allocation without money can be disciplined by artificial currencies [11, 12, 9], with field evidence from food banks [17]. Our “always claim the ceiling” outcome is the babbling case those mechanisms are designed to avoid.

Costly and ex post verification. Verification is the classical remedy [20, 2, 15], with repeated-audit learning variants [8]. Those models are static or lack state dynamics and travel. Our audit mechanisms are simple heuristics in this spirit, not implementations of those optimal mechanisms.

Tolls. Naor [16] shows that individually optimal joining exceeds the social optimum and that a toll equal to the externality restores it [see also 13]. A request price increasing in distance, $\theta + \lambda d$, is the analogous device here; we treat it as a numerical test, not a derivation.

3 Model

Geometry and dynamics. N explorers sit at fixed posts on a ring of circumference 100; one actuator moves at speed 1. Explorer i has health $h_i \in [0, 1]$ and urgency $\tau_i = 1 - h_i$. Its base decay rate is $r_i = r e^{\varepsilon_i}$, $\varepsilon_i \sim \mathcal{N}(0, \text{spread}^2)$, where r is the *load* parameter. Under the *legacy* model the decay is deterministic and linear, so h_i is a function of the time since last service and private information is empty. Under the *v2* (“shock”) model a fraction $(1 - f)$ of the mean decay is deterministic and the rest arrives as hidden Poisson shocks (rate 0.01 per step, exponential size; f is the shock fraction). Health is clipped at 0. Service takes 5 steps (except where noted), is not interruptible, and restores $h_i = 1$. The *v2* configuration in the code additionally uses persistent report noise (below), keeps a commitment when the caller set empties, and records precision at dispatch.

Reports. Explorer i claims $\tau_i + \text{noise} + b_i$ with noise standard deviation $\sigma = 0.15$ (i.i.d. under the legacy model; stationary AR(1) with $\varphi = 0.9$ and marginal sd 0.15 under v2, which is what every v2 result here uses), and $b_i \geq 0$ is its strategic inflation. Requesting is free. The dispatcher observes true h_i only on arrival, and learns each r_i from such observations.

Dispatch rules. **none:** blind spatial round-robin (ignores reports; the polling-style baseline). **greedy_true:** argmax of true urgency, a myopic rule that ignores travel (not an optimal oracle). **learned_index:** ignores claims,

estimates urgency from age and learned rates and serves $\arg \max \min(1, \hat{\tau}_i + \hat{r}_i d_i) / (d_i + \text{service})$. **honest_index**: the same index applied to claims. **fused_index**: Bayesian fusion of the age prior with the claim, then the same index. **audit_index**: **fused_index** plus an ex post audit on arrival that estimates and removes each robot’s claim bias and widens its measurement variance. **audit_disp**: audit applied to the claim at the moment of dispatch (the claim of the robot just chosen is compared with what is later seen). **priced**: honest reporting, but a request is admitted only if $\tau_i > \theta + \lambda d_i$. **dp_step_full**, **dp_age**: the exact benchmarks of Section 4. Unless stated (the **pre** arms of C1 and C7) the simulated policies commit to a target until arrival; only **dp_step_full** re-plans every step. Service is not interruptible.

Metric. Long-run mean team health after a 500-step burn-in (all experiments except C1, which uses 1500 steps without burn-in); secondary metrics are the fraction of robot-steps spent dead ($h = 0$) and the 5th percentile of per-step minimum health. All comparisons are paired on identical seeds (posts, rates, shocks, noise); intervals are 95% t intervals over seeds unless said otherwise. The reference implementation is in numpy; a float64 GPU re-implementation reproduces the simulator seed by seed (maximum absolute difference $\leq 1.2 \times 10^{-15}$ in the stored checks; step DP versus semi-MDP on integer grids $\leq 4 \times 10^{-10}$; GPU versus CPU age-only DP up to 2.7×10^{-5}).

4 Exact benchmarks

Per-step MDP. The state is (actuator position, health grid for each robot); the decision is taken *every step* (including mid-route re-targeting), solved by relative value iteration on a $K+1$ health grid with mean-preserving interpolation and closed-form shock transitions. J_{step}^* (full) is the optimum with free perfect state observation. An earlier semi-MDP that decides only when a service completes yields only the best “commit until arrival” policy; re-planning each step is worth 0.006–0.011 team-health units under shocks (32 seeds, $N=3$, $r=.002$ –.016), so the per-step solution is the reference throughout.

Age-only information. $J^*(\text{age})$ is the optimum for a dispatcher that knows rates and the age of each robot but not h ; the belief is a function of ages so this is an ordinary MDP, exact up to an age cap. Under linear damage age determines h , hence VoI = 0 in the legacy model by construction; we do *not* count this as a finding.

Definition. $\text{VoI}_{\text{step}} = J_{\text{step}}^*(\text{full}) - J^*(\text{age})$.

K-extrapolation and estimator bias. Grid error is one-sided and roughly $1/K$, so values at $K \in \{30, 60\}$ are extrapolated linearly in $1/K$. The estimator is nevertheless biased: in the legacy model, where the theoretical VoI is 0, it returns -0.0014 to -0.0025 . This is comparable to the margins we discuss and is the main reason we do not claim H1 (Section 8).

5 Results: the value of information

C1: correcting the first version. Cell: $N=8$, v_2 , shock fraction 0.5, spread 0.6, service 5, 1500 steps, 32 seeds (CPU reference), loads $r \in \{.0005, .001, .002, .004, .007, .012, .02, .04\}$; paired differences against round-robin (**none**) in team health, whose own mean falls from 0.956 to 0.173 over this range. **greedy_true** is worse than round-robin at every load, with and without preemption (e.g. -0.078 [$-0.092, -0.063$] at $r=.002$ and -0.140 [$-0.159, -0.120$] at $r=.004$ without preemption). The distance-aware **index_true** without preemption loses significantly at $r \in \{.0005, .004, .007\}$ ($-0.004, -0.015, -0.016$), is indistinguishable from round-robin at .001, .002, .012, and wins only at $r \geq 0.02$ ($+0.015$ [$0.003, 0.028$] at .02; $+0.027$ [$0.016, 0.038$] at .04); **learned_index** and **honest_index** without preemption win significantly only at $r=.04$ ($+0.014, +0.027$). Exception: with preemption **index_true** also beats round-robin at $r=.001, .002, .012$ ($+0.005, +0.010, +0.014$). The earlier statement that information is valueless was therefore mostly an artifact of ignoring distance. This matches the DTRP lesson above and is a check, not a discovery.

C2: exact VoI_{step} . In the cell $N=3$, $f=0.5$, spread 0.6, service 5, K -extrapolated from (30, 60), 32 seeds:

load r	.002	.004	.008	.016
VoI_{step}	0.0120 [0.0104, 0.0135]	0.0188 [0.0170, 0.0206]	0.0232 [0.0213, 0.0251]	0.0181 [0.0154, 0.0207]

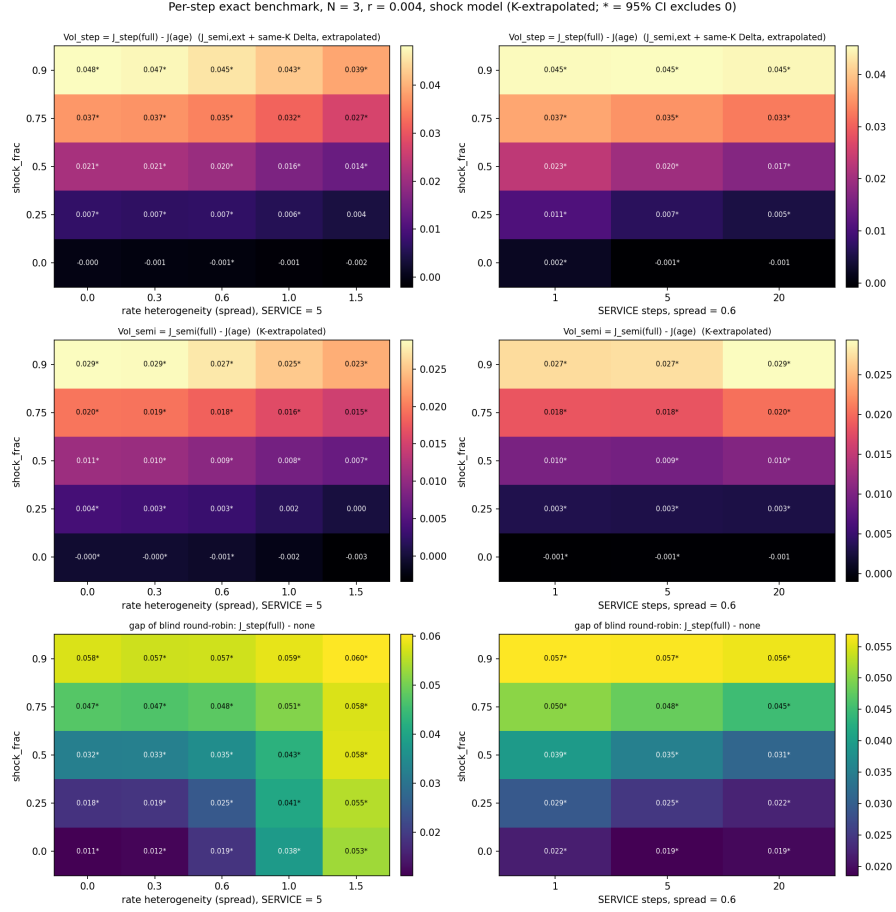


Figure 1: VoI_{step} regions, $N=3$, $r=.004$, $K=(20,30)$, 16 seeds.

This is about twice the value for the non-re-targetable DP (0.0059/0.0096/0.0120/0.0100). The region map (Figure 1; $N=3$, $r=.004$, $K=(20,30)$, 16 seeds) shows VoI_{step} increasing monotonically in the shock fraction (spread 0.6, service 5: -0.003 at $f=0$ (estimator bias), 0.006 at $f=0.25$, 0.020 at 0.5 , 0.035 at 0.75 , 0.045 at $f=0.9$) and, for $f \geq 0.25$, decreasing in spread (at $f=0.5$: 0.021 at spread 0 to 0.014 at 1.5).

H1 is not claimed. The pre-specified claim ($\text{VoI} < 0.02$ at $r=.004$, $f=0.5$) is not supported: at the pre-specified cell the point estimate 0.019 [$0.017, 0.021$] is indistinguishable from the threshold; at service = 1 the region-map estimate (0.024 [$0.020, 0.028$], $K=(20,30)$, 16 seeds) is indistinguishable from the threshold, not a refutation: the same setting at shock fraction 0, where the theoretical VoI is 0, yields $+0.0021$ [$0.001, 0.003$] (`dp_step.json`, region block), and after subtracting this bias the CI lower bound falls below 0.02; and the estimator bias (≈ 0.002) exceeds the margin. Nor can we claim the opposite.

C3: gap of round-robin. The reference is the per-step optimum with free perfect information, so the gap includes VoI. The median gap of none is $0.021/0.038/0.057/0.082$ ($r=.002/.004/.008/.016$; means $0.024/0.042/0.058/0.088$; $v2$, $f=0.5$, spread 0.6, $K=(30,60)$, 32 seeds, 20,000 evaluated steps) at $N=3$, and $0.039/0.058$ ($r=.004/.008$; means $0.044/0.068$; $K=(15,20,30)$, 16 seeds) at $N=4$ (Figure 2; details in `results/dp_step.json`, keys `gap_table.v2` and `n4`). At $N=4$ the exact VoI_{step} is 0.0205 [$0.0181, 0.0229$] and 0.0197 [$0.0168, 0.0225$].

6 Strategic reporting and audits

Game. Explorer i 's utility is its own mean health; its action is a policy for b_i drawn from a finite family, capped at $b \leq 0.8$: *const* (constant b), *timing* (inflate only while the actuator is undecided), *dist10* and *dist30* (inflate

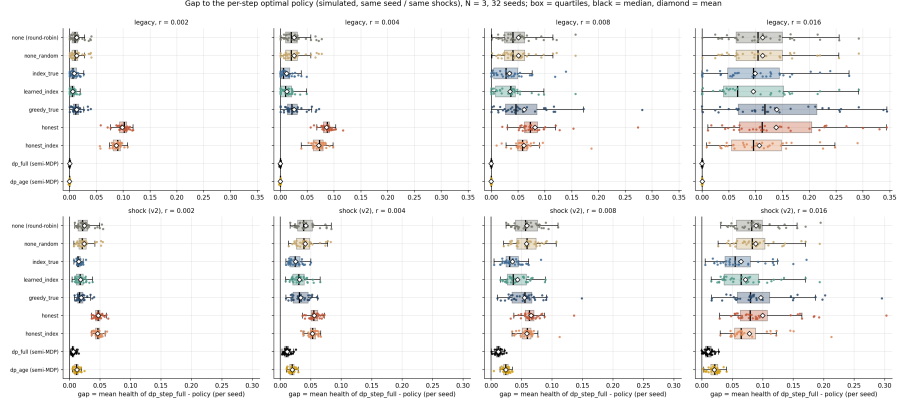


Figure 2: Per-seed distribution of each policy’s gap to the per-step DP (distributions, not only means, are shown because gaps are skewed).

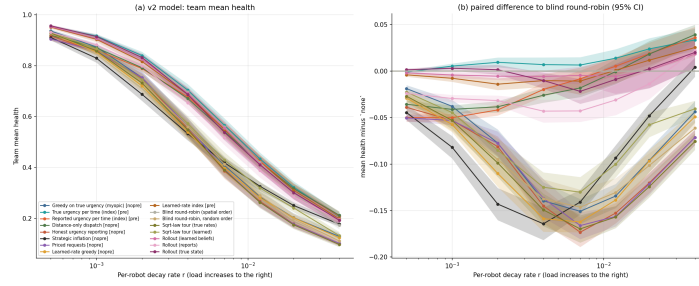


Figure 3: Policies versus load under v2 ($N=8$), supporting C1.

only while the actuator is farther than 10, respectively 30), *taulow* (inflate only while own true urgency is below 0.3), *habit* (inflate only while the actuator is committed), and, for penalty variants only, *adapt* (largest constant inflation that stays under the audit flag threshold), plus truthful reporting. The full design (57 actions, $N=3$, $r=.004$, $f=.5$) has truthful, 16 constants (0.05–0.8) and the five non-adaptive families at eight sizes (0.1–0.8); the reduced design (24 actions) has truthful, 8 constants and the five families at sizes 0.2, 0.4, 0.8. We search for pure-strategy equilibria by best-response iteration with common random numbers, per instance, using a move threshold $\epsilon = 0.002$ (sensitivity at 0.001, 0.005 was pre-specified and run only in the $N=3$ $r=.004$ $f=.5$ and $N=8$ cells). A profile is called an *equilibrium* only when iteration converged; otherwise we say *approximate* (Table 1 gives convergence rates). Final profiles are re-evaluated on fresh shock and noise streams. All game cells use holdout seeds 1000–1031 (32 seeds), 4000 steps at $N=3$ and 2000 at $N=8$, 500 burn-in; penalty parameters were tuned on seeds 0–15 in the first cell only.

Loss. $\text{loss} = \text{health}(\text{truthful}) - \text{health}(\text{approximate best-response profile})$ for the same mechanism, in team-health units, paired over seeds. (Exception: H2 compares the fused equilibrium to *learned_index*; this inconsistency of baselines is disclosed in Section 9.)

C4: naive trust (H2 holds). The loss of *fused_index* is in Table 1. Its equilibrium health is below that of *learned_index*, which ignores claims, in every cell run, with CIs excluding 0. The absolute losses (0.19–0.25 at $N=3$, $f=.5$, against exact VoI_{step} of 0.012–0.023) are the primary evidence; as a side remark the ratio is 17, 13, 8 at $r=.002, .004, .008$ ($f=.5$), and 34 and 8 at $f=.25$ and $.75$ ($r=.004$, using the region-map VoI), so it depends strongly on the cell. The loss is bounded by the inflation cap $b \leq 0.8$. In the exploratory scaling study (Section 7; load calibrated per N : $r = .0098/.0032/.0022$, 25 actions, 4000 steps, 32 seeds) the loss is 0.165/0.405/0.413 at $N = 3/8/16$ (convergence 1.00/0.88/0.56): it rises from 3 to 8 and then saturates. This is the cheap-talk collapse: free claims carry no credible information and informed rules should ignore them; the contribution is the size of the loss relative to the information being protected.

C5: audit at arrival fails (H3 holds). Inflation that depends on distance (inflating when the distance exceeds 10) bypasses the audit at arrival. Table 1 gives the loss; the dead-robot fraction rises from 0.029 to 0.067 at

Table 1: Strategic loss (truthful minus approximate best response), 95% CI over seeds. Convergence rate is the fraction of instances in which best response converged. “Actions” is the size of the strategy set in that cell (the nine $N=3$ cells: 57 at $r=.004, f=.5$; 24 at the other $f=.5$ cells; at $f=.25/.75$ the top three families per mechanism with 24-action sizes). All rows are the same cell and the same game (`full|truth|eps=0.002`, 32 seeds) for the three mechanisms, so the audit rows are directly comparable; every entry is read from `results/game.json`, key `game_v2`. The $N=3$ nine-cell row has no “not run” entries; $N=4$ was not run.

mechanism	cell	loss [95% CI]	conv.	actions
<code>fused_index</code>	$N=3, r=.004, f=.5$	0.247 [0.223, 0.272]	0.875	57
<code>fused_index</code>	$N=3$, nine cells	0.112–0.270	0.875–1.00	57 / 24
<code>fused_index</code>	$N=8, r=.002, f=.5$	0.40 [0.36, 0.44]	0.875	24
<code>audit_index</code>	$N=3, r=.004, f=.5$	0.024 [0.010, 0.038]	0.625	57
<code>audit_index</code>	$N=8, r=.002, f=.5$	0.042 [0.025, 0.059]	0.312	24
<code>audit_disp</code>	$N=3, r=.004, f=.5$	0.0027 [0.0001, 0.0052]	0.250	57
<code>audit_disp</code>	$N=8, r=.002, f=.5$	0.0058 [0.002, 0.010]	0.000	24

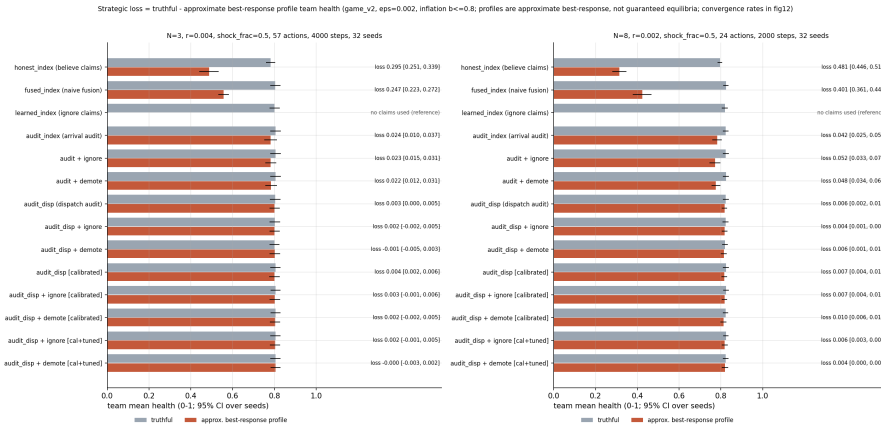


Figure 4: Truthful versus approximate best-response profile for each mechanism (v2 game cells).

$N=3$. Convergence rates are 0.62 ($N=3$) and 0.31 ($N=8$), so the profiles are “approximate”. The CI includes zero in one of the ten cells, $N=3, r=.008, f=.25$ (0.011 [−0.003, 0.025]).

C6: audit at dispatch (H4 refuted and withdrawn). The pre-specified claim was that `audit_disp` is robust within the tested family (loss CI contains 0 and max gain $\leq 2\epsilon$). Both criteria fail: the CI lower bound is > 0 at both N (Table 1) and the maximum one-robot deviation gain exceeds 2ϵ . **We withdraw H4.** What we can claim is descriptive. `audit_index` and `audit_disp` are evaluated in the same cells and the same game (`full|truth|eps=0.002`), where the dispatch-time loss is 1/8.9 ($N=3$) and 1/7.2 ($N=8$) of the arrival-time loss; across all ten cells this ratio ranges from 4.6 to 48, because the denominators are near zero, so we do not state it as a general factor. This ratio should also be read as optimistic for the dispatch audit: convergence is low, and a search that has not converged may fail to find profitable deviations, so the dispatch-audit loss may be underestimated. By contrast, the arrival audit clearly fails (CI excludes 0 in 9 of 10 cells). The evidence on the dispatch audit is inconsistent. In 5 of the 10 cells the loss CI contains zero (these CIs are not corrected for multiple comparisons); in the two cells of Table 1 the CI lower bound is > 0 . At $N=3, r=.004, f=.5$ it contains zero at $\epsilon = 0.005$ (0.0018 [−0.0010, 0.0045]); the calibrated variant has 0.0041 [0.0016, 0.0065] there. Disclosures: (i) max gain is an in-sample quantity. Stored out-of-sample checks exist only for the $N=8$ cell (calibrated mechanisms): the naive in-sample gain is 0.075 for `audit_disp|cal` and 0.100 for `audit_index`, the out-of-sample maximum player gain only 0.0072 and 0.0038 (penalty variants up to 0.0178), consistent with a winner’s curse; (ii) the convergence rate is 0.25 at $N=3$ and 0.00 at $N=8$, so these profiles are loose approximations; (iii) the penalty variants flag truthful robots: the fraction of robots ever flagged under truthful play is 0.61–0.64 at $N=3$ and 0.30–0.34 at $N=8$ before calibration (0.35–0.40 and 0.07–0.11 after).

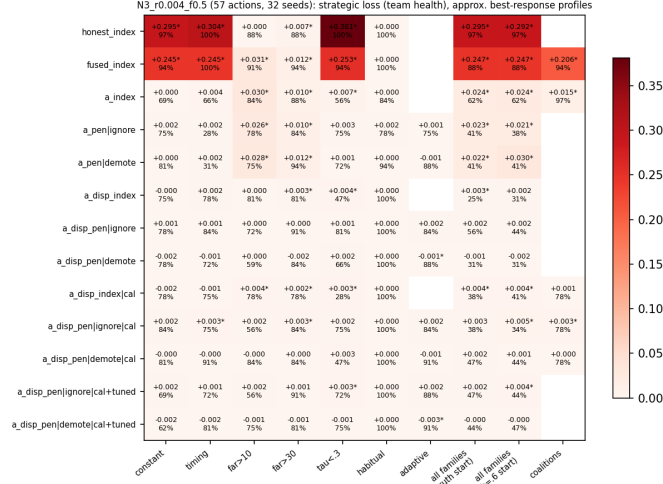


Figure 5: Loss from each strategy family; distance-dependent inflation is what defeats the arrival audit.

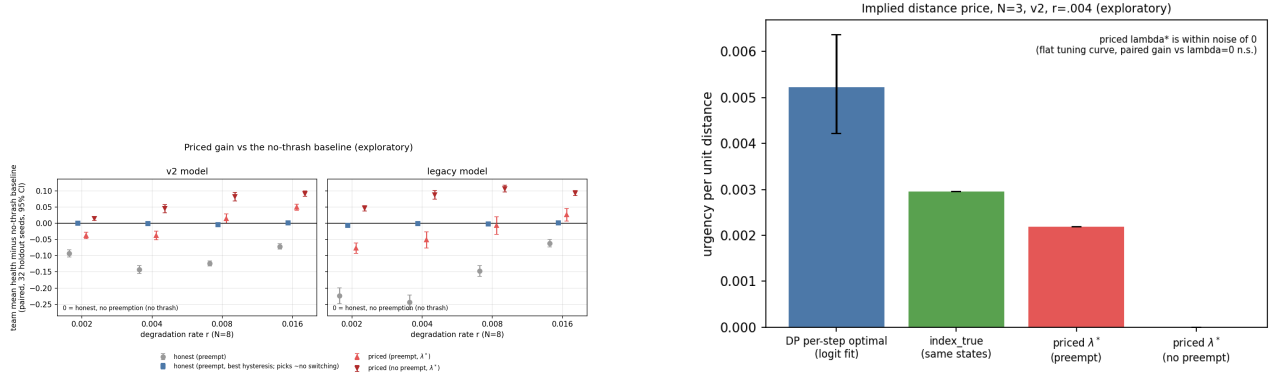


Figure 6: Exploratory (C). Left: priced versus the no-thrash baseline ($N=8$, v2, holdout). Right: implied distance exchange rate of the exact per-step policy.

7 Exploratory: pricing and scale

Everything in this section was run after the pre-specified protocol and is *exploratory*.

C7: pricing. At $N=8$ under v2, λ^* was chosen separately for each load on seeds 0–15 and evaluated on holdout seeds 1000–1031 (6500 steps, spread 0.6). **priced** (no preemption) improves on the no-thrash baseline (honest reporting, no preemption, commitment kept) by +0.015/ +0.045/ +0.082/ +0.092 at $r = .002/.004/.008/.016$, but it remains 0.03–0.08 below **honest_index** (0.029/0.068/0.077/0.052) and 0.06–0.09 below **learned_index** (differences of means). Hence λd is a coarse distance filter. At $N=3$, $r=.004$, $\lambda^* = 0$ without preemption (so **priced** is identical to **honest**) and 0.0022 with preemption (priced minus honest -0.002 [$-0.007, 0.003$]). A fitted conditional-logit index $\tau/(d+5)^{1.14}$ (exponent CI [1.07, 1.21]) predicts the exact per-step policy’s choice with top-1 accuracy 0.77 over 16,552 decisions ($N=3$, $r=.004$, $K=30$, 32 seeds; the argmax-urgency rule predicts 0.51) (Figure 6, right). The hypothesis that λd is a decentralised implementation of the optimal policy is not supported. This is consistent with the Naor picture only qualitatively; we derive no closed form for λ^* . The earlier claim that most of the pricing gain was thrash suppression is supported: at $r=.004$ the no-thrash baseline alone explains 126% (legacy) and 135% (v2) of the gain of preemptive **priced** over preemptive **honest**; with tuned λ^* the net gain over the no-thrash baseline is as above.

C8: scale. With the load calibrated separately per N , round-robin beats every tested information-based heuristic for $N \geq 8$. A cheap value-of-information proxy (a heuristic’s gain over **learned_index**) recovers only

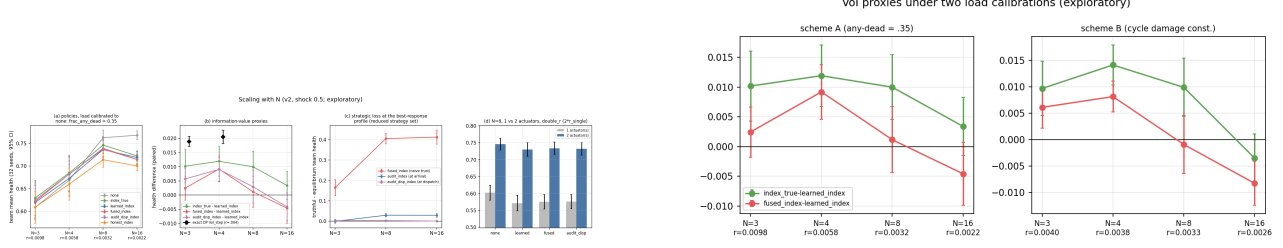


Figure 7: Exploratory (D). Left: policies versus N . Right: VoI proxy under two load calibration schemes.

−5% to +41% of the exact DP value at the six exact cells (it is a severe underestimate and per-seed correlations are −0.08 to 0.59), so true VoI for $N > 4$ is unknown. Round-robin beats all tested heuristics at $N = 8$ and 16 under both load calibrations (e.g. 0.763 against 0.746 for the best heuristic at $N=8$; 0.768 against 0.722 at $N=16$). A two-actuator variant is uninformative: with $A=1$ the fused minus learned difference is itself not significant (Figure 7).

8 Pre-specified protocol and deviations

Nature of the pre-specification. The protocol was written in the middle of the project, after the stage-A and the first stage-B exploratory results. The choice of cells for H3 and H4 was influenced by those results. The document has no external timestamp; it is an internal protocol only. Readers should therefore treat it as an *analysis plan written before the final confirmatory runs*, not as a pre-registration in the strict sense. It was written on 2026-10-02 before the final runs; earlier experiments and everything after are exploratory. Verdicts:

- **H1** (VoI < 0.02 at $N=3$, $r=.004$, $f=0.5$; falsified if CI lower bound ≥ 0.02): *not claimed*. The estimate 0.019 [0.017, 0.021] cannot be separated from the threshold; indistinguishable from the threshold at service = 1 (0.024 [0.020, 0.028]; the same setting at shock fraction 0 gives +0.0021 [0.001, 0.003] where the theory says 0, and after subtracting this bias the lower bound is below 0.02); estimator bias ≈ 0.002 .
- **H2** (fused equilibrium health below **learned_index**; falsified if CI contains 0): *holds* in all cells run (CI excludes 0).
- **H3** (arrival audit has positive equilibrium loss; falsified if both N have CI containing 0): *holds* (0.024 at $N=3$, 0.042 at $N=8$).
- **H4** (dispatch audit loss CI contains 0 and max gain $\leq 2\epsilon$): *refuted per the pre-specified criterion and withdrawn*.

Deviations, all listed here: (1) Of 27 pre-specified cells ($f \in \{.25, .5, .75\}$, $N \in \{3, 4, 8\}$, $r \in \{.002, .004, .008\}$) only 10 were run: nine at $N=3$ and one at $N=8$ ($r=.002$, $f=.5$); $N=4$ is entirely missing. (2) Reduced strategy sets and ϵ sets: the full design (57 actions, three ϵ , coalitions) ran only at $N=3$, $r=.004$, $f=.5$; $N=8$ used 24 actions with three ϵ and coalitions; the other $f=.5$ cells used 24 actions and $\epsilon = 0.002$ only; the $f=.25/.75$ cells ($N=3$ only) used the top three families per mechanism chosen from the $f=.5$ cells and $\epsilon = 0.002$ only (source: **game_v2.notes**). (3) Penalty tuning was done in a single cell. (4) The two DP-region maps use fewer seeds and coarser K than the headline VoI cell. (5) The max-gain criterion is in-sample (Section 6). (6) The loss baseline differs between hypotheses: H2 against **learned_index**; H3, H4 against each mechanism’s own truthful profile. (7) An earlier figure of the strategic results used superseded numbers and was redrawn. Language rules were followed: “equilibrium” only where converged, and “robust” only as “within the tested family”.

9 Limitations

- DP is limited to $N \leq 4$; any statement about $N=8$ or 16 rests on heuristics and a VoI proxy that tracks the DP only loosely.

- Strategy families are finite and inflation is capped at 0.8; a larger family could raise any reported loss, including that of dispatch audits.
- 10 of 27 grid cells were run, $N=4$ none, $N=8$ one cell. Tuning was done in one cell. $\sigma=0.15$, shock rate, and $\varphi=0.9$ each take a single value.
- Health is clipped at 0 and approximate equilibria show a high dead fraction, so losses may largely reflect robots dying.
- Service cannot be interrupted; the model is a ring with fixed posts, not a real robot system.
- Loss baselines are not uniform across hypotheses, and several equilibrium profiles are only approximate (convergence as low as 0.31 for the arrival audit and 0.00 for the $N=8$ dispatch audit).
- Exploratory results (C7, C8) must not be read as confirmatory.

10 Conclusion

In a stylised ring with hidden shocks, the exact value of perfect state information for a travel-coupled dispatcher is of the order of 0.01–0.02 team-health units at $N=3$ and rises with the shock fraction. It is dwarfed by the loss from trusting free claims, which a distance-dependent inflation strategy also exploits against audits at arrival. Auditing at dispatch has a smaller loss in the tested families (about $1/7$ – $1/9$ of the arrival audit in the same cell), but the evidence is inconsistent (CI contains 0 in 5 of 10 cells, uncorrected; convergence 0.25 and 0.00), and the ratio is an optimistic estimate because a non-converged search may miss deviations. Two earlier claims were revised: “information is worthless” mostly reflected ignoring distance, and request pricing is mostly a coarse distance filter. The honest scope is $N \leq 4$ exact, a reduced grid, and finite strategy families.

Reproduction. Code and results are in the repository root. The CPU reference (`arbitration/model.py`, `dp.py`, `game.py`) is verified seed by seed against the GPU code (`arbitration/gpu/`); all headline experiments were run on one GPU (float64 torch). Scripts: `scripts/run_dp_step.py` $\rightarrow$ `results/dp_step.json` (C2, C3), `scripts/run_game_gpu.py` $\rightarrow$ `results/game.json`, key `game_v2` (C4–C6; the older `game_sweep` is exploratory), `scripts/run_c_pricing.py` $\rightarrow$ `results/c_pricing.json`, `scripts/run_d_scaling.py` $\rightarrow$ `results/d_scaling.json`; C1 comes from `results/results.json`. Figures are regenerated by the scripts’ `figs` stages and `python -m arbitration.figures`. Interactive demo: `docs/demo/index.html`.

References

- [1] Manjari Asawa and Demosthenis Teneketzis. Multi-armed bandits with switching penalties. *IEEE Transactions on Automatic Control*, 41(3):328–348, 1996.
- [2] Elchanan Ben-Porath, Eddie Dekel, and Barton L. Lipman. Optimal allocation with costly verification. *American Economic Review*, 104(12):3779–3813, 2014.
- [3] Dimitris J. Bertsimas and Garrett van Ryzin. A stochastic and dynamic vehicle routing problem in the Euclidean plane. *Operations Research*, 39(4):601–615, 1991.
- [4] Dimitris J. Bertsimas and Garrett van Ryzin. Stochastic and dynamic vehicle routing with general interarrival and service time distributions. *Advances in Applied Probability*, 25(4):947–978, 1993.
- [5] Onno J. Boxma and W. P. Groenendijk. Pseudo-conservation laws in cyclic-service systems. *Journal of Applied Probability*, 24:949–964, 1987.
- [6] Francesco Bullo, Emilio Frazzoli, Marco Pavone, Ketan Savla, and Stephen L. Smith. Dynamic vehicle routing for robotic systems. *Proceedings of the IEEE*, 99(9):1482–1504, 2011.
- [7] Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50(6):1431–1451, 1982.

- [8] Yan Dai, Moise Blanchard, and Patrick Jaillet. Non-monetary mechanism design without priors: achieving efficiency via adaptive costly audits, 2025. A preliminary version appeared at COLT 2025 (38th Annual Conference on Learning Theory).
- [9] Ezzat Elokda, Saverio Bolognani, Andrea Censi, Florian Dörfler, and Emilio Frazzoli. A self-contained karma economy for the dynamic allocation of common resources. *Dynamic Games and Applications*, 14(3): 578–610, 2024.
- [10] Kevin D. Glazebrook, Diego Ruiz-Hernandez, and Christopher Kirkbride. Some indexable families of restless bandit problems. *Advances in Applied Probability*, 38(3):643–672, 2006.
- [11] Artur Gorokh, Siddhartha Banerjee, and Krishnamurthy Iyer. From monetary to non-monetary mechanism design via artificial currencies. *Mathematics of Operations Research*, 46(3):835–855, 2021.
- [12] Mingyu Guo, Vincent Conitzer, and Daniel Reeves. Competitive repeated allocation without payments. In *WINE 2009, LNCS 5929*, 2009.
- [13] Refael Hassin and Moshe Haviv. *To Queue or Not to Queue: Equilibrium Behavior in Queueing Systems*. Kluwer/Springer, 2003.
- [14] Jerome Le Ny, Munther Dahleh, and Eric Feron. A linear programming relaxation and a heuristic for the restless bandit problem with general switching costs, 2008.
- [15] Tymofiy Mylovanov and Andriy Zapechelnuk. Optimal allocation with ex post verification and limited penalties. *American Economic Review*, 107(9):2666–2694, 2017.
- [16] Pinhas Naor. The regulation of queue size by levying tolls. *Econometrica*, 37(1):15–24, 1969.
- [17] Canice Prendergast. The allocation of food to food banks. *Journal of Political Economy*, 130(8), 2022.
- [18] Hideaki Takagi. Queuing analysis of polling models. *ACM Computing Surveys*, 20(1):5–28, 1988.
- [19] Dongnuan Tian and Rob Shone. Dynamic repair and maintenance of heterogeneous machines dispersed on a network: a rollout method for online reinforcement learning, 2026.
- [20] Robert M. Townsend. Optimal contracts and competitive markets with costly state verification. *Journal of Economic Theory*, 21(2):265–293, 1979.
- [21] S. S. Villar. Indexability and optimal index policies for a class of reinitialising restless bandits. *Probability in the Engineering and Informational Sciences*, 30(1):1–23, 2016.
- [22] Richard R. Weber and Gideon Weiss. On an index policy for restless bandits. *Journal of Applied Probability*, 27(3):637–648, 1990.
- [23] Peter Whittle. Restless bandits: activity allocation in a changing world. *Journal of Applied Probability*, 25 (A):287–298, 1988.